

# Mental Health Sentiment Analysis

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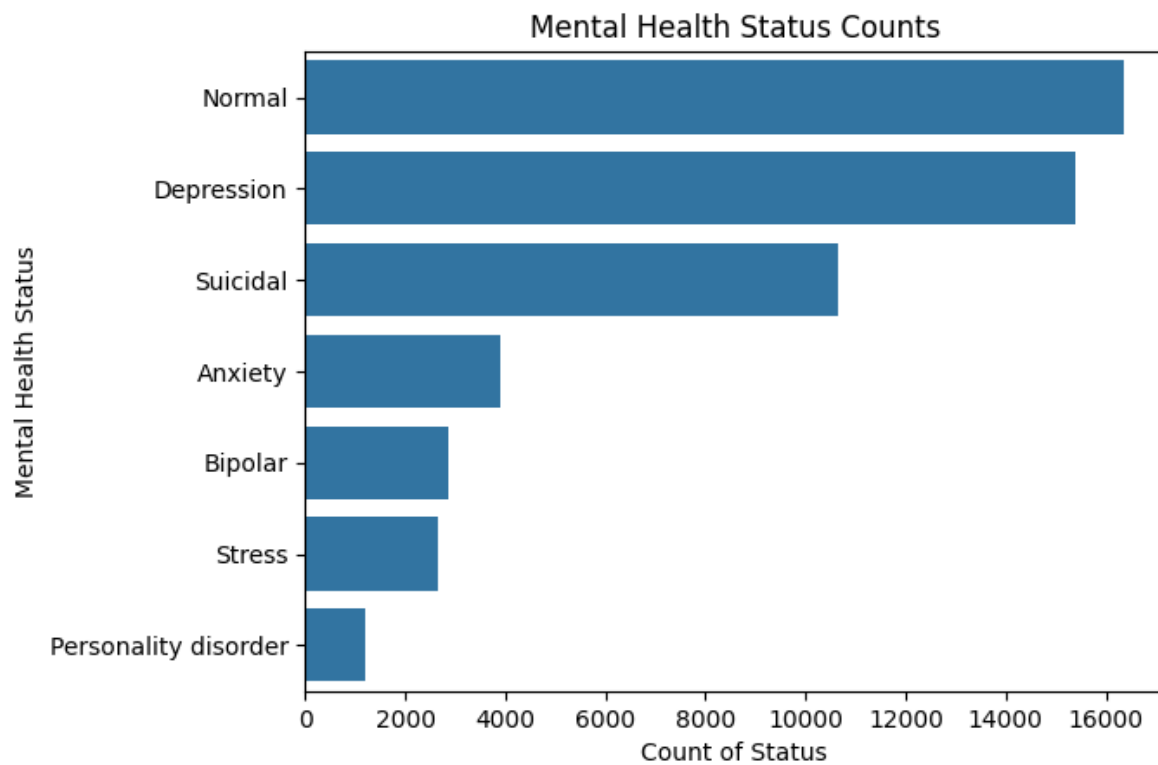
# Introduction

- Globally, mental health is a growing concern in the United States and around the world.
- One in five U.S. adults experience mental illness each year and one in six U.S. youth, ages 6-17, experience mental health disorders each year.
- Suicide is the 2<sup>nd</sup> leading cause of death among people ages 10-14.
- With early and proper detection these deaths could be prevented and people at risk could get the help they need.
- Project objectives:
  - Familiarize ourselves with the various mental health classes and identify patterns.
  - Visualize our data to help with modeling assumptions.
  - Accurately predict the mental health statuses for each statement.

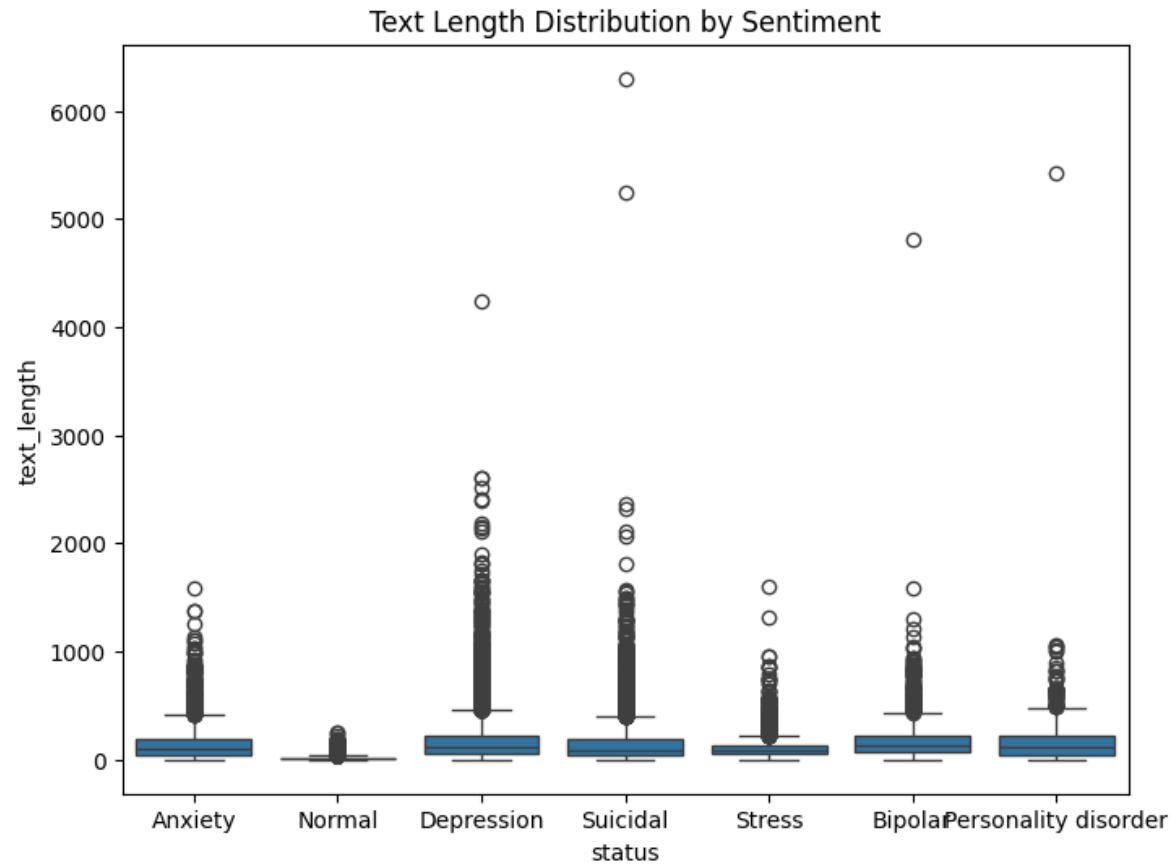
# Data Description

- Our data contains 53,042 rows and three columns.
- Each row contains a mental health statement where each row represents a unique individual comment.

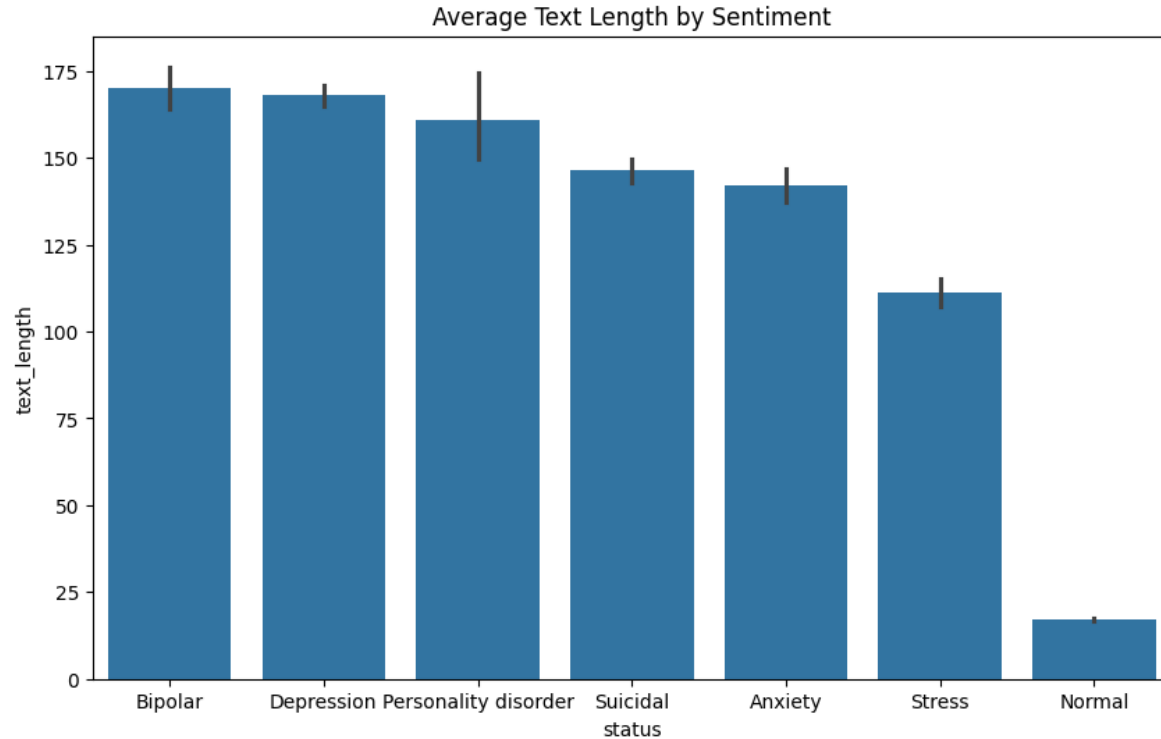
| Column name | Type   | Description                                                                                         |
|-------------|--------|-----------------------------------------------------------------------------------------------------|
| Unique_id   | int    | Unique identifier for each statement.                                                               |
| Statement   | String | Mental health statement.                                                                            |
| Status      | String | Mental health status (Normal, Stress, Anxiety, Suicidal, Personality disorder, Depression, Bipolar) |



- Most cases are Normal, Depression, and Suicidal.

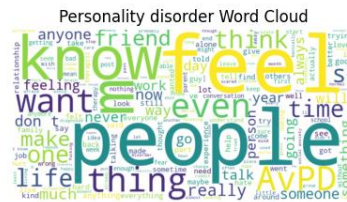
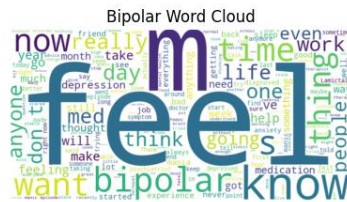
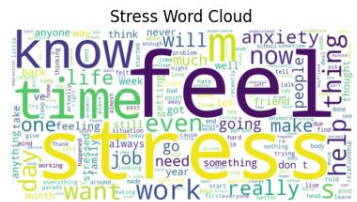
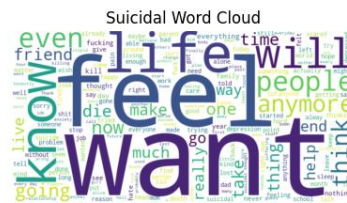
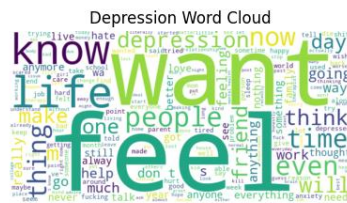
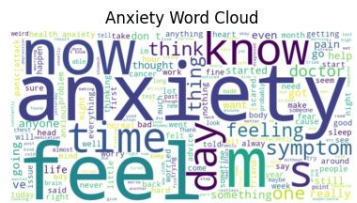


- There are some outlier statements with exceptionally long text lengths.
- The majority of statements are below 3,000.
- The Normal class has the shortest text lengths overall.



- Normal statements have the least average text length.
- Statements with mental health problems tend to have longer statements on average.

- [illegible]



- These clouds do not give much insight.
- The main thing to notice from all the classes with mental health issues is the words "feel", "know", and the words of their classification showing up in the clouds.



# Baseline Modeling

- Used zero-shot-classification from transformers.
- Default model, facebook/bart-large-mnli.
- Received an accuracy of 38%.

```
# Initialize pipeline
classifier = pipeline('zero-shot-classification', model='facebook/bart-large-mnli', device=0)

# Define candidate labels
candidate_labels = ['Normal', 'Depression', 'Suicidal', 'Anxiety', 'Bipolar', 'Stress', 'Personality disorder']

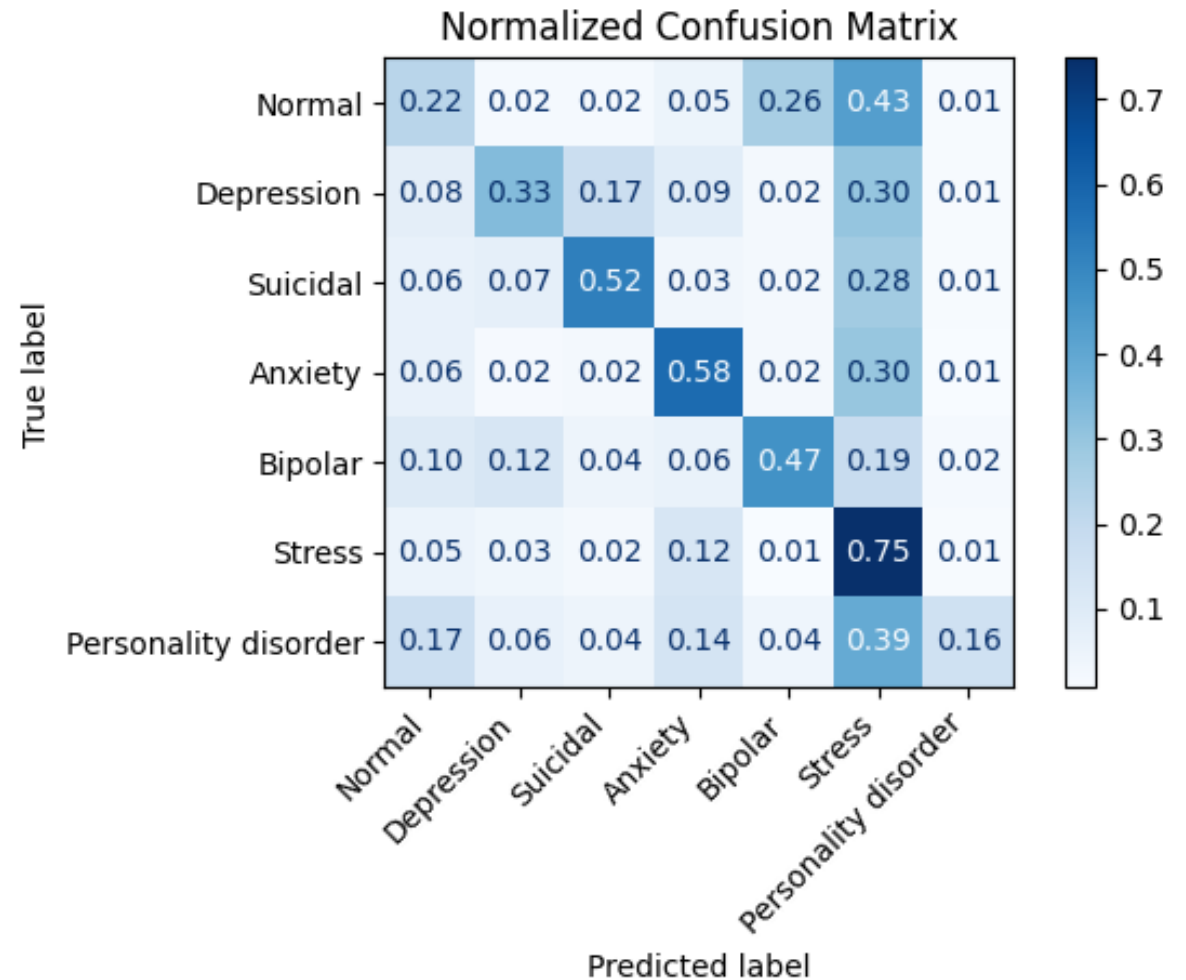
# Run classification in batch
def classify(batch):
    results = classifier(batch['statement'], candidate_labels=candidate_labels)
    return {'status_label': [r['labels'][0] for r in results]}

# Apply function
dataset = dataset.map(classify, batched=True, batch_size=16)

# Convert back to DataFrame
df = dataset.to_pandas()

# Evaluate
from sklearn.metrics import accuracy_score
print("Accuracy:", accuracy_score(df['status'], df['status_label']))
```

- The results of this were similar for each model we tested.
- Model predicted most of the statements to be Stress even though it is the second to last common label.
- This model also struggled to identify the largest Normal class.



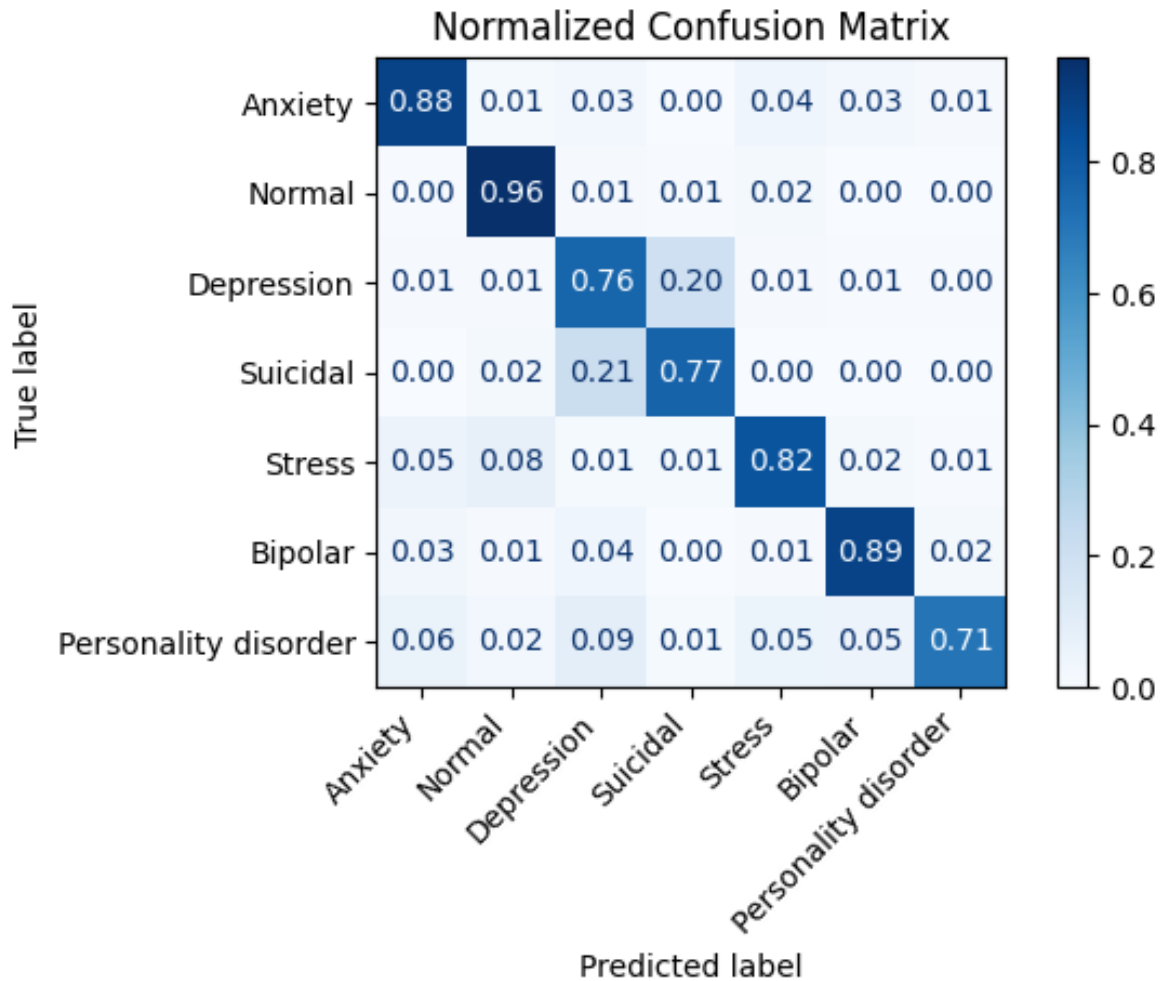
```
# Training arguments
training_args = TrainingArguments(
    output_dir="./mental-health-model",
    evaluation_strategy="epoch",
    save_strategy="epoch",
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=4,
    weight_decay=0.01,
    logging_dir='./logs',
    load_best_model_at_end=True,
    metric_for_best_model='accuracy'
)
```

- nateraw/bert-base-uncased-emotion from Hugging Face.
- Used 4 epochs and batch size 16.
- Showed signs of overfitting on the validation set.

---

## Final Model

```
# Tokenizer from the mental health BERT model
model_name = "nateraw/bert-base-uncased-emotion"
tokenizer = AutoTokenizer.from_pretrained(model_name)
```



- Shows a significant improvement over our baseline predicting across all mental health classes.
- The model had a slight problem differentiating between the Depression and Suicidal classes.

# Conclusion

- Mental health is growing concern in our country and across the world.
- Through our exploratory data analysis, we were able to point out the most common mental health conditions in our data as well as common traits in the statements given by the individuals in our data.
- Our baseline modeling did not give us favorable results in identifying mental health conditions in the text statements in our dataset.
- We received much better results when we used a model pre-trained on mental health data and fine-tuned for our specific use case.
- Schools, workplaces, or healthcare systems are some potential industries that could benefit from a system like this to give students, employees, or patients support.
- We did have some potential overfitting in our final model so in further study we could work on fine tuning this model to perform better on the validation set.